NASA SPACE APPS CHALLENGE 2026 / SYLHET TEAM

BHUSANKET

Earth Systems Discovery & Evidence Lab

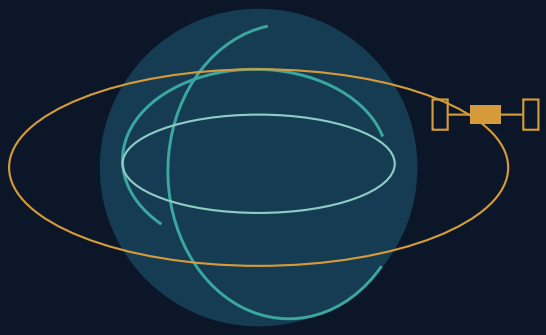
Read Earth’s signals. Trace every change.

NASA DATA

OBSERVE / TEST / UNDERSTAND

FROM A QUESTION

TO TRACEABLE EVIDENCE



MASTER PRODUCT, UX & BUILD STRATEGY

A complete blueprint for a distinctive, student-friendly and research-useful software prototype built for “Be an Earth System Trend Detective!”

THE PRODUCT PROMISE

Anyone can start with a real question about Earth, investigate it with NASA data, see a scientifically defensible trend result, and understand how strong the evidence is — in Bangla or English.

Team size 4–6 members	Selected challenge Be an Earth System Trend Detective!
First geography Bangladesh-wide pilot	Product scope Bangladesh-first, globally reusable

Working brand name. Final naming, domain and trademark availability are not yet verified.

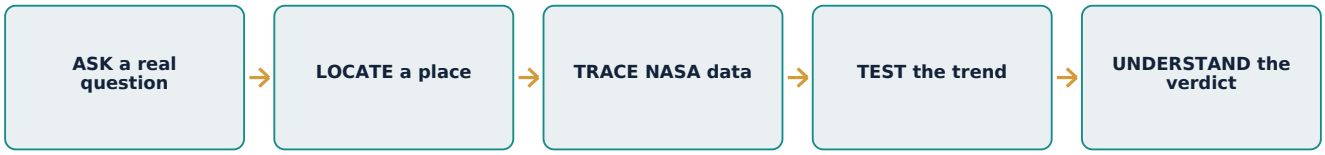
01 / THE DECISION

One product. One shared scientific engine. Three ways to use it.

This plan consolidates the earlier TrendSleuth and ClimaClue concepts, the mentor build-guide direction shared by the team, the selected NASA challenge and the Bangladesh Participants Guide. The goal is not to build a generic climate dashboard or a collection of disconnected demos.

BhuSanket is a question-first investigation platform. A user chooses a place and asks whether a measurable Earth-system variable is changing. The platform retrieves a compatible NASA dataset, creates a time series, performs a reproducible statistical test and explains the result with its uncertainty and source history.

The central user journey



The product’s five defining principles

• Question-first: begin with “Is rainfall changing here?” rather than a technical parameter catalogue.

• Evidence-first: every chart and verdict is tied to a named NASA product, variable, units, dates and processing steps.

• Honest uncertainty: “no significant trend” and “not enough evidence” are valid outcomes, not failures.

• Learning by doing: students and public users learn to test claims rather than simply consume charts.

• Bangladesh-first, not Bangladesh-only: start with a national case library; keep the data workflow location-independent.

THE X-FACTOR

Claim Court turns the same scientific engine into a short evidence-investigation game. The Analysis-Window Slider lets users see whether a claim depends on a cherry-picked time period. A regional comparison reveals that one variable can trend differently in different places.

What success looks like

A first-time student can complete one real investigation without help; a researcher can reproduce the result from its Data Passport; and a judge can understand the product’s value in under ten seconds and see the working evidence trail in the live demo.

Important: this strategy improves the quality of the entry but cannot guarantee a local award, global nomination or Global Award.

02 / CHALLENGE FIT

Map the product directly to the NASA challenge and score sheet.

The selected challenge asks teams to identify Earth-science variables, acquire regional data, visualize sufficiently long time series, quantify change and test statistical significance. BhuSanket is designed around these exact requirements.

Challenge requirement	BhuSanket response	Proof visible to a judge
Identify variables of interest	Question-to-variable catalogue: temperature, precipitation, vegetation, soil moisture and selected atmospheric variables.	Question cards map to a clearly defined variable, unit and source.
Acquire regional data	NASA adapters with location selection; point series first, gridded area aggregation when validated.	Data Passport names product, version, location, period and retrieval mode.
Visualize a long time series	Interactive chart with full record, missing-data marks, units, trend line and selected analysis window.	A real NASA-derived series is visible and downloadable.
Quantify and test the trend	Mann–Kendall significance, Theil–Sen slope and confidence interval with quality checks.	Direction, change per decade, p-value, CI and caution status.
Make the result understandable	Plain-language verdict, Bangla/English explanations, Student Lab and Claim Court.	A first-time user can interpret the result without knowing the test names.

Judging score sheet — from the Bangladesh Participants Guide

Criterion	Points	What our prototype must demonstrate
Impact	1–20	Who benefits, how far the solution can reach, and a credible way to measure usefulness.
Creativity	1–20	A fresh interaction and execution: claim investigation, window sensitivity and regional contrast.
Validity	1–20	Real data, tested statistics, honest limitations, technical feasibility and usability.
Relevance	1–20	NASA data is essential to the product, not a logo or decorative map layer.
Presentation	1–20	A clear beginning, middle and end; a memorable live demonstration and polished visuals.
Teamwork	1–5	Visible member roles, commits, issues, decisions and genuine collaboration.
User experience	1–5	A beginner can use it without significant training.
NASA data usage	1–5	NASA and, where appropriate, partner/third-party sources are clearly shown and credited.
Challenge / repo / page	0–3	Correct challenge category, public repository and complete submitted project page.

SCORING STRATEGY

The guide gives 20 points each to Impact, Creativity, Validity, Relevance and Presentation; smaller points cover teamwork, user experience, NASA data usage and submission completeness. Build evidence for each criterion. Do not assume that a beautiful UI compensates for weak science.

Source basis: user-provided Space Apps Challenge Participants Guide, judging section, pp. 13–17. The guide lists a maximum of 118 points before any stated participation bonus; confirm all current rules with the Local Lead.

03 / PRODUCT ARCHITECTURE

Three workspaces — one shared engine.

The platform should feel powerful without looking complicated. The user sees three clear paths. All paths use the same data, computation and evidence layer so results stay consistent.

Workspace	Who it serves	Primary job	What it contains
Explore Earth	Public, journalists, curious citizens	Answer a real-life question about a selected place.	Question-first map, variable cards, time series, simple verdict, comparison and evidence trail.
Student Lab	School and university learners, teachers	Learn to investigate environmental evidence.	Guided missions, hypothesis prompt, chart-reading lessons, Claim Court, glossary, reflection and classroom cases.
Research Workspace	Researchers, postgraduate students, advanced users	Run reproducible trend investigations.	Custom variable, region, time period, aggregation, dataset, quality filters, statistics and exports.

The navigation should stay short

Top-level navigation	Purpose
Explore	Default entry point for place- and question-led investigations.
Student Lab	Guided learning and missions.
Claim Court	The memorable evidence game; reachable from Explore and Student Lab.
Research	Advanced controls and export.
Data & Methods	NASA product catalogue, statistical method notes, limitations and provenance.

Avoid separate, disconnected dashboards. A student should be able to open a case in Explore, switch to Student Lab to learn the method, and open the same case in Research to inspect the full metadata.

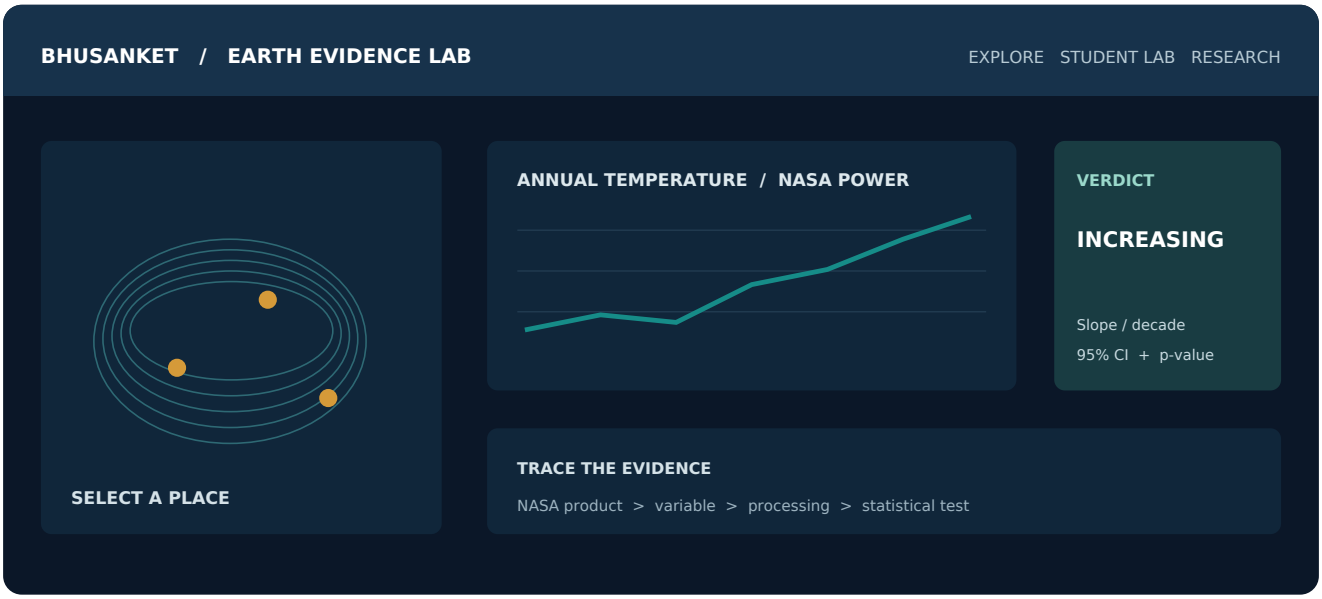
PROGRESSIVE DISCLOSURE

Show the question, chart and plain-language verdict first. Put p-values, test assumptions, aggregation choices and quality details behind clearly labelled “Evidence & methods” controls. Never force a beginner to learn Mann–Kendall before they can start.

04 / VISUAL IDENTITY & UX

Make it memorable — not a generic AI dashboard.

The design direction is a modern scientific field notebook mixed with a calm mission-control instrument panel. Real maps, real chart geometry and evidence labels should create the visual identity. Avoid a template-heavy AI look.



Visual system

Design token	Recommended value	Use
Deep ink	#132238 / #0B1728	Navigation, dark workspace, strong contrast.
Signal teal	#168C88	Primary actions, selected states, evidence paths.
Warm paper	#F6F5F0	Reading surfaces, case notes, learning content.
Evidence amber	#D59A39	Caution labels, uncertainty, important evidence markers.
Alert red	#B9574E	Data-quality warnings only; never use as decoration.

Distinctive visual motifs

• Evidence-case labels: numbered case IDs, compact source tags and “case closed” stamps.

• Contour-line / orbital mark: a subtle brand motif connecting Earth mapping and satellite observation.

• Mission-to-evidence trail: a visual path from satellite/product to variable, processing, test and verdict.

• Real data as visual material: maps, time-series traces, uncertainty bands and missing-data markers — not random stock space imagery.

• Purposeful motion: slider recomputes results, map selection updates the chart, and the verdict stamp appears only after analysis completes.

Interaction rules

• One obvious primary action per screen.

• Use plain labels such as “Choose a place” and “Test this claim”; keep advanced settings in a drawer.

• Do not rely on colour alone: every verdict has a word, icon and explanatory text.

• Use readable chart labels, keyboard-friendly controls, mobile-responsive layout and accessible contrast.

• Avoid excessive gradients, glowing cards, decorative animations, crowded metric tiles and generic chatbot bubbles.

THE 10-SECOND TEST

Show the home screen to someone who has never heard the pitch. Within ten seconds they should say: “This tool checks how an Earth variable changes over time using NASA data.” If they cannot, simplify the page.

Typography: use one clean sans-serif family (e.g., Inter, Manrope or Source Sans) with two weights. Keep all-caps for short labels only.

05 / SCREEN-BY-SCREEN BLUEPRINT

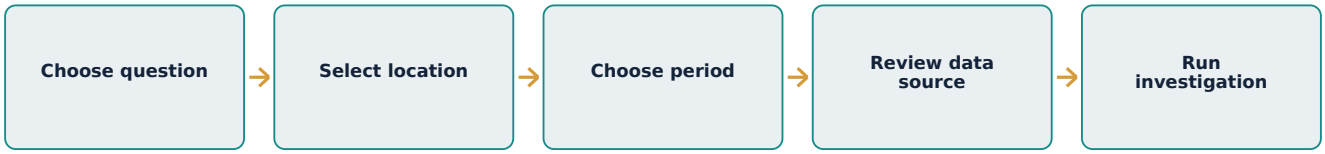
The first visit should feel obvious and inviting.

The homepage is not a data dump. It is an invitation to investigate. A first-time user should start with a question, choose a place, and reach a meaningful result in a few clear steps.

Home screen

- Hero: “Earth is always changing. Are you seeing the real signal?”
- One-sentence explanation: “Explore environmental changes using real NASA Earth-science data.”
- Question tiles: Getting hotter? Rainfall changing? Vegetation changing? Water changing?
- Primary action: “Start an investigation”. Secondary action: “Try a sample case”.
- Three workspace cards: Explore Earth, Student Lab, Research Workspace.
- Featured case: one verified Bangladesh investigation with source and period visible.
- Trust strip: NASA data, transparent methods, Bangla/English, no fabricated results.

Explore flow



Screen	Required components	Key interaction
1. Explore landing	Question cards, location search, Bangladesh map and global map toggle.	Tap a question or select a point/district.
2. Location picker	Search, map, district selector, coordinate option and location context.	Select point or district; app explains whether the value is a point/grid estimate or area average.
3. Investigation setup	Variable, dataset recommendation, start/end dates, annual/seasonal aggregation.	Beginner sees recommended defaults; Research can open advanced settings.
4. Evidence workspace	Map, time-series chart, trend line, selected time window, source chip and comparison button.	Hover/tap a year for exact value; drag the analysis-window slider.
5. Verdict	Trend direction, magnitude, significance, CI, quality state and “What does this mean?”	Open the Evidence & methods drawer or compare another region.

Important first-time behaviour

The app should preselect a sensible variable and time period when a user asks a simple question. It should not ask the user to understand product codes, grid projections or statistical tests before the first result. Advanced choices remain available but optional.

06 / CORE INVESTIGATION EXPERIENCE

The evidence workspace is the heart of the product.

The chart, map and verdict must tell one consistent story. The same prepared series must feed the chart, statistical calculations, comparison and export.

Evidence workspace layout

Area	Content
Top context	Question, selected place, selected period, dataset chip and units.
Map panel	Selected location highlighted; neighbouring region available for comparison; boundary and grid context explained.
Chart panel	Observed annual/seasonal values, missing periods, Theil–Sen line and selected-window overlay.
Verdict panel	Increasing / decreasing / no significant trend / not enough evidence / window-sensitive.
Evidence drawer	Dataset ID, product version, variable, source URL, retrieval date, aggregation, QC, missingness and method.
Actions	Compare another place, change analysis window, download case report, open Student Lab.

Verdict card — mandatory fields

Field	Example format (illustrative only)
Direction	Increasing / decreasing / no significant trend detected
Magnitude	Theil–Sen slope: +X °C per decade
Statistical evidence	Mann–Kendall p-value; 95% confidence interval
Coverage	Number of valid years / total period; missing percentage
Reliability state	Evidence sufficient / caution / insufficient
Interpretation	One short sentence in plain English or Bangla
Limitations	Grid scale, source type, seasonal choice, autocorrelation or data gaps.

TWO DIFFERENT “NO” STATES

“No significant trend detected” means the test did not find sufficient evidence of a monotonic trend in the selected record. “Not enough evidence” means coverage, record length, data quality or unresolved assumptions prevent a defensible verdict. These must never be merged into one vague status.

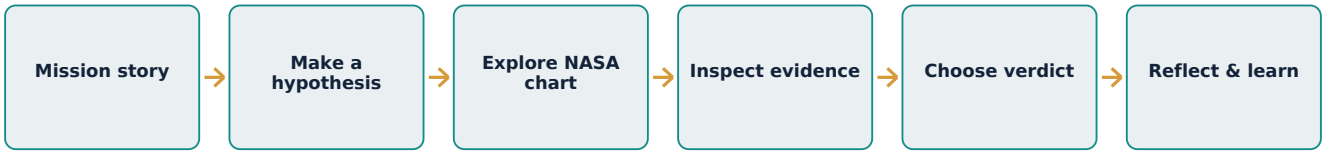
Do not say that a place is “safe”, “unsafe”, “doomed”, “flood-prone next year” or “caused by climate change” based on a trend test alone. Trend detection is not forecasting or causal attribution.

07 / STUDENT LAB

Turn Earth science into a guided investigation.

Students should not feel they are opening a research instrument panel. They should feel they are solving a small scientific mystery with real data.

A guided mission has six steps



Student feature	How it works	Learning outcome
Mission cards	Short real-world prompts such as “Is the monsoon changing in this region?”	Connect science to place and lived experience.
Hypothesis prompt	Student predicts increasing, decreasing or uncertain before seeing the verdict.	Understand that science tests expectations.
Chart-reading coach	Small tips for axis, units, period, missing values and trend line.	Read a graph rather than just look at it.
Plain-language glossary	Trend, slope, p-value, confidence interval, seasonality and grid cell in Bangla/English.	Reduce technical barriers.
Evidence reflection	Student writes why they selected a verdict and what evidence changed their mind.	Practice reasoning and scientific humility.
Teacher pack	Five prepared case worksheets and discussion questions.	Make the product usable in a classroom.

Student experience levels

Level	Case design	Unlock condition
Rookie	Clear trend, clean chart, one variable and one place.	Correctly identify direction and basic meaning.
Detective	Noisy data, seasonal pattern or missing years.	Inspect evidence and explain uncertainty.
Chief Detective	Window-sensitive claim, opposite regional patterns or data-quality trap.	Use multiple evidence tools and justify a cautious verdict.

Student Lab should include a pre/post chart-reading quiz. Do not claim educational impact until actual users complete it and results are recorded.

08 / SIGNATURE EXPERIENCE

Claim Court — the feature judges should remember.

Claim Court is not a separate toy. It is a playable version of the same scientific investigation engine used by Explore and Research. Every chart and score is generated from the actual selected NASA data.

The round



Evidence tool	What it reveals
Full record	The complete available time series, not just the short interval quoted in a claim.
Season check	Like-for-like seasonal totals or a validated seasonal treatment.
Trend test	Slope, p-value, 95% confidence interval and quality warnings.
Neighbour check	The same variable in a second location to test whether the pattern is regional.

Verdict choices

Stamp	Meaning
SUPPORTED	The full-record analysis supports the claimed direction under the defined variable, location and period.
WINDOW-DEPENDENT	A selected sub-period tells a materially different story from the full record.
NOT SUPPORTED ON THIS RECORD	The full record does not support the claimed direction.
NOT ENOUGH EVIDENCE	The record is too short, gappy, low quality or blocked by unresolved statistical assumptions.

Scoring rules

Suggested game scoring: +3 for a correct final verdict; +1 for each unspent evidence token; at least one tool must be used; −2 for confidently stamping a misleading claim as SUPPORTED. Add a confidence slider: a confident wrong answer should cost more than an appropriately cautious answer. Keep the score explanation visible.

Pilot cases

• Dhaka summer temperature claim — test a defined summer air-temperature metric and full record.

• Rajshahi annual rainfall claim — define “drier” explicitly as a change in annual rainfall total; do not imply drought without a separate definition.

• Khulna monsoon rainfall claim — compare like-for-like monsoon totals.

• Add Sylhet/Sunamganj as a case only after verifying data availability and choosing a defensible variable.

CASE SELECTION RULE

These are candidate questions, not pre-announced findings. Run the analysis first, then choose cases that provide a balanced mix of supported, window-dependent, unsupported and insufficient-evidence outcomes. Never write the verdict in advance.

09 / RESEARCH WORKSPACE

Give researchers control without overwhelming everyone else.

Research Workspace is the advanced path, not the default landing page. It should make analysis reproducible and transparent rather than adding complexity to the public experience.

Control	Expected behaviour
Variable catalogue	Searchable variables grouped by land, atmosphere, water and vegetation; each includes unit, source and meaning.
Dataset picker	Only show products compatible with the variable, period, region and selected temporal aggregation.
Geography	Point, grid cell or polygon selection; show the actual spatial support and valid-pixel coverage.
Time range	Display the actual available dates; warn about short records and gaps.
Aggregation	Annual, calendar month, like-for-like season or scientifically appropriate seasonal method.
Quality controls	Allow inspection of QC flags, fill values, missingness and excluded observations.
Statistical settings	Mann–Kendall variant, autocorrelation handling, confidence level and multiple-testing adjustment where applicable.
Comparison	Compare the same variable and compatible product across locations; do not compare unlike measurements.
Export	CSV time series, JSON result, chart image and a reproducibility report / PDF.

Research output package

- Raw or source-linked data reference and the processed time series.
- A Data Passport with collection version, parameter, units, period and spatial context.
- Aggregation and quality-control decisions.
- Statistical test, slope, confidence interval, p-value and assumptions.
- Code / app version and retrieval timestamp.
- A short interpretation that distinguishes result from hypothesis or causal explanation.

REPRODUCIBILITY TEST

A researcher should be able to take the exported Data Passport and processed series, repeat the same method and recover the same numerical result within the documented tolerance.

10 / SIGNATURE FEATURES

Six memorable experiences, delivered in a controlled order.

The product can have a rich identity without having dozens of disconnected features. The first two signature moments must be polished; the remaining ones strengthen the product when the core is stable.

Feature	User-visible experience	Build priority	Why it matters
1. Analysis-Window Slider	Drag start/end years; the app recomputes the selected-period trend and keeps the full-record result beside it.	P1 — first WOW	Makes cherry-picking and time-window sensitivity visible in seconds.
2. Trend Divergence Detector	Same variable, two places; shows increasing/decreasing/no-significant-trend side by side.	P1 — first WOW	Directly demonstrates that the same Earth process can produce different regional signals.
3. Climate Claim Checker	Choose a preset claim; app maps it to place, variable, season and period, then tests it.	P1	Practical public and journalism use; avoids an unreliable free-text AI verdict.
4. Claim Court	Three evidence tokens, evidence tools, prediction, confidence and verdict stamp.	P1 — signature	Turns statistics into a short learning game using real data.
5. NASA Mission-to-Evidence Trail	Visual path from satellite / model product to variable, processing, test and verdict.	P1/P2	Shows judges exactly how NASA data powers the solution.
6. Earth Impact Lens	Carefully worded context about why a trend may matter for agriculture, water, education or planning.	P2	Connects scientific results to human relevance without claiming causation.

Optional extensions

• Hot-days counter: count days above a user-defined, clearly stated temperature threshold per year; validate daily data and threshold choice before release.

• Shareable Case Card: export one-page evidence summary with chart, verdict, source and caveats.

• Daily Case: one verified case per day after the core product is stable.

• AI variable helper: suggests candidate variables from a user question but asks the user to confirm; it does not choose the statistical answer.

Do not build in the first release: forecasting, causal attribution, a general-purpose climate chatbot, a large social network, complex multi-agent orchestration or every NASA dataset.

11 / NASA DATA STRATEGY

Start with a small, defensible data portfolio.

The mentor guide and team notes identify NASA POWER, MODIS/VIIRS, GPM IMERG and GRACE as useful data directions. The wider catalogue below adds other NASA Earth-science products. Every product must pass a feasibility and quality check before it is enabled in the interface.

Tier 0 — first working pipeline

Dataset / product	What it can contribute	Use in BhuSanket	Caution
NASA POWER Daily / Monthly	Meteorological time series: T2M, T2M_MAX, PRECTOTCORR and selected solar / humidity parameters.	First pipeline for temperature and precipitation questions; point series and cached daily/monthly data.	Analysis-ready gridded / model-derived data. Show product parameter, units, time standard and spatial support; do not label it a raw satellite measurement.

Tier 1 — first satellite-based extensions

Dataset / product	Variable	Use	Resolution / notes
GPM IMERG V07B	Precipitation rate / accumulation	Rainfall and monsoon trend; regional grid and comparison.	About 0.1° / 10 km; 30-minute source data; daily/monthly aggregation possible; retrospective Final Run is preferable for research comparisons.
Terra MODIS MOD11A2.061	Day/night land-surface temperature	Surface heating patterns and day/night contrast.	8-day composite, 1 km; use QC layers and scale factor; LST is not 2-m air temperature.
Terra MODIS MOD13Q1.061	NDVI / EVI vegetation indices	Vegetation greenness trends.	16-day, 250 m product family; apply QA and scale factor; check the current collection specification before coding.
VIIRS land products	LST / vegetation / fire, depending on selected product	Higher-frequency or complementary satellite view.	Select an exact product/version first; verify temporal coverage, QA and comparability with MODIS.

DATA TYPE MUST BE VISIBLE

POWER and MERRA-2 are model/reanalysis-based data products; IMERG is a multi-satellite precipitation estimate; MODIS and VIIRS products are satellite-derived observations or retrievals. The interface must identify the source type accurately.

Recommended initial sequence: POWER temperature + precipitation first; then one GPM IMERG rainfall case; then one MODIS LST or NDVI case. Do not promise that all three will be live until the ingestion and quality-control pipelines pass.

12 / EXPANDED NASA CATALOGUE

Add more NASA missions only when the question justifies them.

These datasets make the product expandable beyond temperature and rainfall. They should appear as a Research Workspace catalogue with clear “available”, “planned” or “not yet validated” status.

Dataset / mission	Possible variables	Potential investigation	Key limitation / implementation note
SMAP	Surface and root-zone soil moisture	Soil moisture change and land-surface water availability.	Daily products exist at different resolutions; select one collection and QC scheme. Record is shorter than POWER.
GRACE / GRACE-FO	Terrestrial water storage anomalies	Large-basin or regional-scale water storage trends.	Monthly and coarse spatial support; not a local well-level groundwater measurement. Basin-scale interpretation is safer.
MERRA-2	Atmospheric temperature, humidity, wind, aerosols and other reanalysis fields	Long-term atmospheric conditions and selected Earth-system variables.	Reanalysis, not a direct single-satellite sensor product; grid is coarse relative to a city or district.
GLDAS	Soil moisture, evapotranspiration, runoff, surface temperature and water storage variables	Land-surface water and energy balance investigations.	Model output; verify product version, land-surface model, forcing and temporal frequency.
AIRS / CLIMCAPS	Atmospheric temperature and water-vapour profiles; trace gases	Atmospheric structure and moisture changes.	Vertical profile / retrieval data need careful pressure-level and QC handling; not equivalent to near-surface temperature.
OMI / Aura	Ozone, NO2, SO2, aerosol-related products	Atmospheric composition and air-quality-related trend studies.	Column or swath measurements; choose a specific gridded product and handle coverage / quality flags.

Optional contextual / validation sources

Where access, licensing and quality metadata allow, BhuSanket may add Bangladesh Meteorological Department station data, Bangladesh Water Development Board hydrological records or other trusted agency observations as independent context or validation. These sources must be clearly labelled as non-NASA and must not replace the NASA data at the centre of the challenge response.

Dataset onboarding checklist

- Exact collection short name and version confirmed.
- Variable / band, units, scale factor, fill value and QA fields understood.
- Temporal coverage and resolution checked for the requested period.
- Spatial resolution and grid support displayed accurately.
- Data access method, credentials and redistribution terms checked.
- One small-area sample file successfully opened and plotted.
- Missingness and valid-pixel coverage computed.
- A repeatable adapter and provenance record added before the UI exposes the dataset.

13 / GEOGRAPHIC STRATEGY

Bangladesh is our first case library — not our boundary.

The user competes from Sylhet, but the software should not be locked to Sylhet or to one region. Bangladesh-wide discovery gives the project local relevance; a location-independent data adapter supports global reuse.

Candidate Bangladesh demonstration cases

Location	Candidate question	Possible variable / source	Selection rule
Dhaka	Are summer temperatures changing?	POWER T2M / T2M_MAX; later MODIS LST for surface temperature.	Define season and distinguish air temperature from land-surface temperature.
Rajshahi	Is annual rainfall changing?	POWER precipitation; later GPM IMERG.	Define “drier” as a measured rainfall metric; do not equate it automatically with drought.
Khulna	Is monsoon rainfall changing?	POWER precipitation / GPM IMERG seasonal accumulation.	Use like-for-like monsoon totals and disclose missingness.
Sylhet / Sunamganj	How are rainfall patterns changing?	POWER and GPM IMERG; selected seasonal or annual metrics.	Use verified data and avoid implying a forecast of flood or flash-flood risk.

Point versus district average — a critical rule

• POWER point request: a coordinate or representative point returns a point/grid estimate. Do not label it “the district average”.

• Gridded products: for a district polygon, aggregate valid pixels inside the boundary using a documented method; report the number/percentage of valid pixels and area weighting where appropriate.

• Global case: add one non-Bangladesh location only after verifying the same product, units, period and analysis path. A map that looks global is not proof that every location is supported.

Bangladesh-first, global-ready data architecture



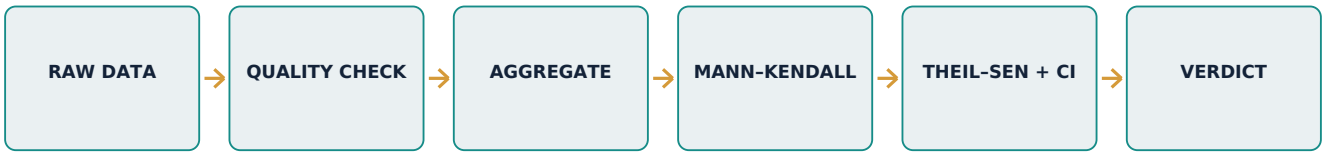
PILOT RULE

Use Dhaka, Rajshahi, Khulna and Sylhet/Sunamganj as candidate demonstration locations. Calculate the trends first; feature only results that pass data and statistical checks. The locations are pilot choices, not pre-claimed scientific findings.

14 / SCIENTIFIC ENGINE

Numbers must be defensible — and repeatable.

The statistical engine is deterministic Python code. It receives a prepared time series and returns a structured result. An AI model may explain that result, but it must never calculate or invent the statistics.



Stage	Implementation rule
1. Ingest	Preserve source file / API response, product version, retrieval timestamp, units and original metadata.
2. Clean	Parse time, apply scale factors and fill values, inspect quality flags, remove invalid observations and record gaps. Never silently interpolate.
3. Aggregate	Temperature: annual mean or mean of daily maxima, explicitly defined. Rainfall: annual or monsoon total. Use like-for-like seasons.
4. Trend test	Mann–Kendall for monotonic trend; use a seasonal method or defensible seasonal aggregation when seasonality matters.
5. Slope	Theil–Sen slope; report change per year or decade in original units and a 95% confidence interval.
6. Dependence	Inspect serial autocorrelation. If material, apply a validated correction (e.g., Hamed–Rao) or withhold the significance verdict.
7. Multiple testing	When scanning many districts / variables, control false discoveries (e.g., Benjamini–Hochberg FDR) and label adjusted results.
8. Validate	Unit-test rising, falling, flat, seasonal, tied, missing and autocorrelated synthetic series; compare outputs with a trusted implementation.

Result states

State	Meaning shown to user
Significant increase / decrease	Evidence supports the direction; show slope, p-value and CI.
No significant trend detected	The test did not find sufficient evidence of a monotonic trend. This does not prove that nothing changed.
Not enough evidence	Coverage, record length, data quality or unresolved assumptions prevent a reliable verdict.
Window-sensitive	The selected interval and full record lead to materially different conclusions.

STATISTICAL SAFETY

Statistical significance is not the same as practical importance. A small change can be significant; a large-looking change can be uncertain. Show magnitude and uncertainty together. Do not claim causation or forecast future conditions from a trend test.

15 / TRUST, AI AND PROVENANCE

Every result needs a traceable NASA Data Passport.

A user or judge should be able to answer: where did this number come from, how was it processed, and what can it legitimately tell us?

Data Passport — required fields

Group	Fields
Identity	Analysis ID; dataset short name; collection / version; NASA DAAC or API; source URL.
Measurement	Variable / band / parameter; unit; scale factor; source type (satellite retrieval, multi-satellite estimate, model or reanalysis).
Space & time	Point coordinates or polygon ID; grid / spatial resolution; temporal resolution; start/end dates; time standard.
Processing	Quality flags used; fill values removed; aggregation rule; seasonal definition; missing count / percentage; valid-pixel coverage.
Statistics	Test variant; slope method; slope per decade; p-value; confidence interval; autocorrelation decision; multiple-test correction.
Reproducibility	Retrieval timestamp; cached/live status; software / code version; assumptions; limitations.

AI boundary

• AI may translate a verified result, explain slope / p-value / confidence interval in simple Bangla or English, and suggest a candidate variable for a user question.

• AI must receive a locked result object. It may not change numerical fields, invent observations, choose the statistical verdict or claim causation.

• The UI must distinguish generated explanation from computed evidence.

• Log the AI tool, model, prompt, input result and output for the project’s documentation and disclosure.

Suggested result object

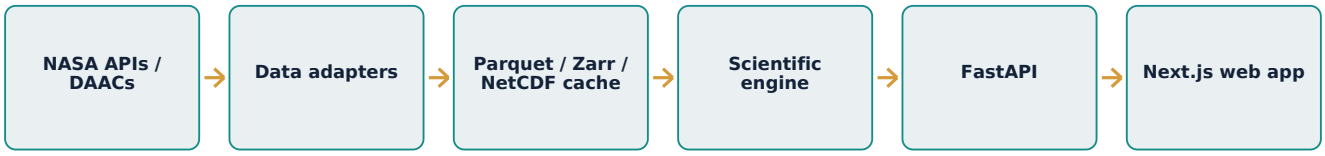
```
{
"analysis_id": "case-001",
"dataset": {"short_name": "POWER_DAILY", "version": "recorded-version"},
"variable": {"id": "T2M", "unit": "degC"},
"location": {"type": "point", "label": "Dhaka", "lat": "...", "lon": "..."},
"period": {"start": "YYYY-MM-DD", "end": "YYYY-MM-DD"},
"aggregation": "annual_mean",
"quality": {"valid_observations": 0, "missing_percent": 0.0},
"trend": {"direction": "computed", "slope_per_decade": 0.0,
"p_value": 1.0, "ci_95": [0.0, 0.0]},
"verdict": "computed_state",
"provenance": {"source_url": "...", "retrieved_at": "...",
"cache_mode": "offline"}
}
```

16 / TECHNICAL ARCHITECTURE

Keep the stack simple, modular and offline-first.

The mentor build-guide direction is preserved: pre-cache the data, keep the statistics engine deterministic, serve results through a small API and make the demo survive network failure.

System flow



Layer	Recommended stack / responsibility
Frontend	Next.js + React + TypeScript; Tailwind CSS; MapLibre or Leaflet; ECharts / Recharts / Plotly for charts.
Backend	Python FastAPI; Pandas, NumPy, SciPy; tested Mann–Kendall implementation and Theil–Sen estimator.
Geospatial	xarray, rasterio, GeoPandas when needed. Start with point time series; add raster/polygon aggregation after the core works.
Storage	Parquet for cached time series and results; Zarr / NetCDF for gridded arrays; JSON for Data Passport metadata.
AI layer	Optional narration / translation service. Receives a verified result object only.
Deployment	Public web demo plus offline mode backed by prepared datasets and deterministic fixtures.

Core modules

- data/ — NASA adapters, API clients, local cache readers and metadata normalizers.
- compute/trend.py — aggregation, Mann–Kendall, Theil–Sen, confidence intervals and quality flags.
- claims/ — preset claims, case definitions and verdict rules.
- api/ — FastAPI routes and stable response schemas.
- explain/ — optional AI narration / translation, isolated from calculations.
- tests/ — statistical, data-quality, API and offline-mode tests.

Suggested API surface

Endpoint	Purpose
GET /variables	Return supported variables, units, compatible datasets and definitions.
GET /locations	Return supported places / polygons and spatial metadata.
GET /claims	Return verified preset claim definitions and case metadata.
POST /trend	Run a single trend investigation.
POST /compare	Compare the same variable across compatible locations.
GET /data-passport/{analysis_id}	Return full provenance and reproducibility details.
POST /explain	Optional narration; accepts only a verified result object.

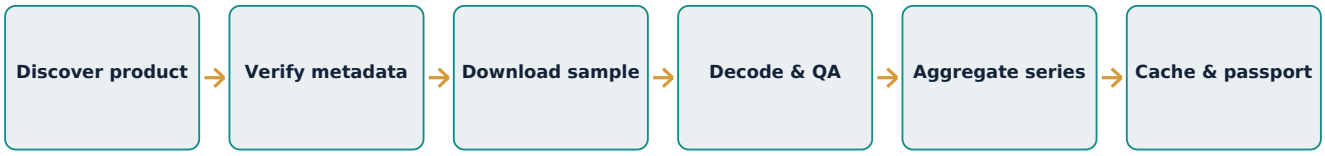
Version the API response schema. Do not let the frontend calculate significance. Do not allow the AI layer to edit numerical fields.

17 / DATA ENGINEERING & OFFLINE RELIABILITY

Prepare data before the demo; never depend on venue internet.

Large Earth-observation files can be slow, credential-gated or difficult to process live. The app should show a live/cached status honestly and use the same result schema in both modes.

Acquisition sequence



Practical implementation

• Create a free NASA Earthdata account for products that require authentication; verify credentials and access before building the adapter.

• Use NASA POWER first for a low-friction data pipeline. Verify exact parameter names, community, units, time standard and response schema.

• Use Earthdata Search / CMR and product documentation to identify exact GPM, MODIS, SMAP, GRACE, AIRS and OMI collections.

• Download only a small sample first. Open it with xarray / h5py / rasterio as appropriate; inspect variable names, dimensions, scale factors, fill values and quality flags.

• Cache cleaned and aggregated demonstration series as Parquet; keep raw product references and source metadata.

• For gridded raster products, store source tiles / subsets as NetCDF or Zarr when appropriate; avoid loading a whole global product into memory.

• Prepare offline fixtures that use the exact same API response schema as the live computation path.

• Show a visible badge: LIVE DATA, CACHED NASA DATA or DEMO FIXTURE. Never imply a fixture is a live query.

OFFLINE-FIRST DEMO RULE

A judge must be able to run the prepared cases without internet. The chart, statistics, verdict and Data Passport must all derive from the same cached time series. Keep one local recorded demo as a fallback, but show the working app first.

Mentor-guide alignment

The shared build-kit screenshots proposed pre-event acquisition, batch/streaming, agentic design, MCP servers and skill files. For this challenge, batch ingestion and local caching are the reliable foundation. Agentic or MCP orchestration is optional; do not add it unless it solves a concrete user task and can be demonstrated without weakening the deterministic science pipeline.

18 / HUMAN IMPACT

Make the usefulness measurable, not just emotional.

BhuSanket should help people learn and investigate, but it must not overstate the real-world consequences of a trend. Its strongest immediate impact is environmental data literacy and faster access to reproducible regional evidence.

Audience	Practical value	How we prove usefulness
Students	Learn to read a time series, distinguish trend direction from significance and investigate a local question in Bangla or English.	Short pre/post chart-reading quiz; record actual score change and task completion time.
Teachers	Use guided missions, case cards and printable worksheets in class.	Teacher reviews one case; record feedback and whether instructions are understandable.
Journalists / public	Check whether a climate statement is supported by a selected NASA record before sharing it.	Track claims tested and whether users revise their first judgement after seeing evidence.
Researchers	Get a reproducible starting point with source, period, units, aggregation and statistics.	Ask a researcher to reproduce one result from the Data Passport.
Community groups / planners	Use regional evidence as a starting point for further assessment.	Expert review; clearly state that trend detection alone is not a policy instruction.

Pilot evaluation plan

• Recruit 5–8 university students, including people who have not studied climate statistics.

• Give a short pre-test on chart reading, trend direction, p-value meaning and data limitations.

• Ask each participant to complete one investigation without help; record time, confusion points and assistance required.

• Give a comparable post-test and report the actual sample size and measured change. Do not invent an impact percentage.

• Ask one teacher, domain researcher or local expert to review a Bangladesh case and its interpretation.

Global impact pathway

Bangladesh is the first case library. The same variable → compatible NASA product → regional time series → trend test → evidence verdict workflow can be reused in other countries. Add a non-Bangladesh case only after its data path is verified. Over time, the product can support climate literacy, research reproducibility and responsible public communication across regions.

19 / TEAM EXECUTION

Give every member ownership — not just a name on the project page.

For a 4–6 person team, divide the work by deliverable. The product/technical lead owns integration and scope. Each member should have visible work in the repository and a clear speaking role.

Role	Ownership
1. Product & technical lead	Scope, architecture, integration, API contracts, technical decisions and final demo flow. Strong fit for the CSE / AI-ML-skilled member.
2. NASA data engineer	POWER parameters, NASA product discovery, acquisition, cache, metadata and dataset adapters.
3. Science / statistics lead	Aggregation rules, Mann–Kendall, Theil–Sen, CI, autocorrelation, FDR and scientific review.
4. Frontend / interaction lead	Map, question flow, chart, slider, verdict card, responsive layout and micro-interactions.
5. UX / education / content lead	Plain-language explanations, Bangla glossary, student missions, Claim Court cases and user tests.
6. QA / video / submission lead	End-to-end tests, public links, documentation, demo recording, video subtitles and NASA project page.

Combining roles

Team size	Practical arrangement
4	Product + science; data + backend; frontend + UX; QA + video + documentation.
5	Separate product/science from data/backend; frontend; UX/education; QA/video/submission.
6	Use all six roles above; appoint one integration owner and hold a daily merge/checkpoint.

Working agreements

- Public GitHub repository from the beginning; meaningful commits and short README updates.
- Use issues for tasks, pull requests for integration and a short decision log.
- Agree on a single API result schema before frontend/backend work diverges.
- Keep a daily 10-minute demo: each person shows one working contribution.
- No one builds a separate feature without agreeing how it connects to the shared result engine.

20 / BUILD ROADMAP

The 48-hour event is the integration and presentation window — not the start of the project.

The team can prepare the data, scientific engine, user experience and case content in advance. During the local hackathon, protect time for integration, usability, recording and submission.

When	Main work	Exit condition
10 Nov — lock & verify	Freeze scope; public repo; confirm POWER parameters; request real data for 3–4 pilot locations; cache first series; agree units and time standard.	A real NASA response is stored with metadata; one plot can be generated.
11 Nov — science core	Implement aggregation, Mann–Kendall, Theil–Sen and CI; add data-quality flags; test synthetic rising/falling/flat/missing series; compare with trusted library.	Statistics pass tests and match trusted implementation within tolerance.
12 Nov — product slice	Build Home → Explore → Result; integrate map, chart, verdict and Data Passport; add window slider; prepare offline cases; run first-time-user test.	One real investigation works end-to-end. Freeze core scope.
13 Nov — Bangladesh event day 1	Polish Claim Court; integrate all modules; rehearse 240-second local pitch; record working demo; name NASA datasets out loud; submit local judging video according to Local Lead schedule.	Stable local demo and video uploaded.
14 Nov — Bangladesh event day 2	Final QA; 30-second global video by the guide’s stated noon cut-off; complete NASA project page, public repo, AI disclosure, NASA data list and references; submit early.	Public links work; project page is complete; final video and documentation are accessible.

Daily working rhythm

Start of day	Midday	Evening	End of day
Agree one milestone and owner.	Integrate and test across the API boundary.	Run a judge-style demo and collect feedback.	Commit, back up data and record unresolved issues.

FEATURE FREEZE

By the evening of 12 November, the core investigation must work. After that, prioritise reliability, accessibility, documentation and presentation. If a feature is unstable, remove it from the live demo instead of showing a broken interaction.

Schedule note: the user-provided Participants Guide has an inconsistency for the Video 2 date in one table. Its run-of-show places the 240-second local judging video at 6:30 PM on 13 November. Confirm the final deadline with the Sylhet Local Lead.

21 / QUALITY GATES & RISK REGISTER

A polished product is a product that does not lie or break.

Judges will test whether the prototype works, whether the numbers are defensible and whether a first-time user can understand it. Use release gates rather than relying on last-minute visual checks.

Risk	Mitigation / release gate
NASA API slow or unavailable	Pre-download and cache; retain local fixtures with the same schema; label LIVE / CACHED / FIXTURE honestly.
Parameter or product mismatch	Verify exact variable, units, version, time standard, scale factor and QA before coding the screen.
Seasonality creates a false story	Use annual or like-for-like seasonal series; document aggregation; use a seasonal method where appropriate.
Autocorrelation inflates significance	Test serial dependence; apply validated correction or withhold significance verdict.
District value misrepresented	Label point/grid values as point/grid; use polygon aggregation only with valid pixels and coverage.
Too many features	Protect P0; add only the slider, one comparison and Claim Court after the end-to-end flow is stable.
AI invents a statistic	AI receives locked result JSON; numerical fields are rendered from backend, never generated by model.
Impact claims exaggerated	Report measured user-test outcomes only; separate plausible applications from proven outcomes.
Demo / public link fails	Test app, repo and videos in a private browser window; keep local recorded fallback.
Video compliance issue	English subtitles; name every member; credit third-party / generated assets; no identifiable under-18 faces, names or voices.

Release checklist — every box must pass

- One real NASA dataset is loaded and its product, variable, units, period and source are visible.
- Chart, slope, p-value and confidence interval are generated from the same series.
- Missing-data and weak-evidence cases are not forced into an increase/decrease verdict.
- Analysis-window slider recomputes the result and compares it with the full record.
- At least one second location can be compared using the same variable and compatible method.
- Claim Court uses the same engine as Explore.
- The app works offline for prepared demo cases.
- A first-time student can finish one investigation without help.
- The Data Passport allows a reviewer to trace and reproduce the result.

22 / PITCH, VIDEOS & SUBMISSION

Make the project easy to understand and easy to judge.

The Bangladesh Participants Guide says teams make two 240-second videos and one 30-second global video. The local judging video must show the working solution and state NASA data usage aloud. The global video must show the problem, solution, demo and NASA data usage.

240-second local judging video — suggested beat sheet

Time	What to say / show
0:00–0:30 — Hook	Open with: “Can a chart make a climate claim look true when the full record tells a different story?” Show the BhuSanket home screen.
0:30–1:00 — Problem	Raw Earth data is powerful, but people struggle to find variables, understand trends and judge significance.
1:00–1:40 — NASA evidence	Select a Bangladesh place and question. Name the NASA dataset and variable aloud. Show the actual source and period.
1:40–2:30 — Live product	Run the trend; show slope, p-value, confidence interval and quality status. Move the Analysis-Window Slider; compare another region.
2:30–3:15 — X-factor	Play one Claim Court round. Let the judge see that the game uses the same scientific engine.
3:15–4:00 — Impact & close	Show Student Lab and Research Workspace. State the real user-test result, limitations and global reuse pathway.

30-second global video — visual beat sheet

Time	Visual / narration
0–5 sec	Team name + BhuSanket identity.
5–11 sec	Problem: environmental trends and claims are difficult to interpret from raw data.
11–22 sec	Show the working app: NASA product → chart → slider → evidence verdict. Name the NASA dataset.
22–30 sec	Impact: students, researchers and the public can investigate a place and understand the evidence. End on brand.

NASA project page — required content

• High-Level Summary: project name, what was built, who it helps, how it works; include the 240-second video and public GitHub link.

• Project Demo: 30-second video (or the guide’s accepted seven-slide alternative).

• Final Project: GitHub, design, documentation and demo materials.

• Project Details: why / what / how; community involvement; tools and technologies.

• Use of AI: name every AI tool; explain prompts, reasoning and data it worked on.

• NASA data sources: list every NASA / partner dataset and exactly what was used.

• References: datasets, papers, libraries, images, fonts, templates and other assets.

PUBLIC ACCESS CHECK

Every repository, video and documentation link must open without login or permission request. Test from a private / incognito browser window. An unsubmitted NASA project page is not judged.

23 / FINAL BUILD CONTRACT

What we commit to deliver — and what we deliberately leave out.

This is the scope contract for the team. A small, complete and trustworthy product is stronger than a wide product full of unfinished buttons.

Must be complete for the competition

• One real NASA-powered investigation works end-to-end.

• A clear question-first entry and Bangladesh-wide location selection work.

• Time-series chart shows units, dates, gaps and trend line.

• Mann–Kendall, Theil–Sen slope and 95% CI are computed by tested code.

• Verdict distinguishes significant trend, no significant trend, insufficient evidence and window sensitivity.

• Data Passport traces the source and processing.

• English and Bangla explanation are available.

• Analysis-Window Slider works on a real case.

• At least one two-location comparison works.

• One Claim Court round uses the same engine.

• Public repository, working demo, videos and NASA project page are complete.

Strong if time permits

• One GPM IMERG or MODIS product integrated and quality-checked.

• Student mission pack with a short pre/post quiz.

• Shareable Case Card / reproducibility PDF.

• One verified non-Bangladesh case to demonstrate global reuse.

• Earth Impact Lens with carefully qualified context.

Explicitly out of scope for this build

• Climate forecasting or disaster early warning.

• Causal attribution or policy recommendations based only on a trend test.

• A general-purpose chatbot that generates environmental conclusions.

• Every NASA dataset, every district polygon and every variable in the first release.

• Complex multi-agent orchestration, MCP servers or user-account/social features unless a concrete tested need emerges.

FINAL PRODUCT STATEMENT

BhuSanket turns NASA Earth-science data into an accessible investigation experience. It helps people discover what is changing, where it is changing, how large the change is and whether the evidence is statistically strong — while teaching them not to confuse a striking chart with a reliable conclusion.

24 / BRAND, REFERENCES & WORKING ASSUMPTIONS

Keep the identity human, credible and easy to remember.

BhuSanket is the recommended working name because it connects a local linguistic identity with the product's core idea: reading signals from Earth. Do a domain and trademark check before final lock.

Name option	Meaning / positioning	Recommendation
BhuSanket	Earth signals; concise, locally rooted and internationally pronounceable.	Recommended working brand.
BhuSutra	Patterns and connections across Earth systems.	Strong research-oriented alternative.
DharaKhoj	Discovering changes in the Earth / natural systems.	More playful and student-friendly.
BhuChinho	Traces or signs of Earth's changes.	Distinctive but slightly less direct.
Niriksho	Observation / investigation-inspired name.	Short, brandable alternative; meaning should be explained.

Tagline options

Read Earth's signals. Trace every change. | Test the claim. Trace the NASA evidence. | Discover the trend. Understand the evidence.

Source references

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[12] NASA AIRS mission and data products
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[14] Hamed, K. H. & Rao, A. R. (1998). A modified Mann–Kendall trend test for autocorrelated data. Journal of Hydrology, 204(1–4), 182–196. DOI: 10.1016/S0022-1694(97)00125-X
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[15] Sen, P. K. (1968). Estimates of the regression coefficient based on Kendall's tau. Journal of the American Statistical Association, 63(324), 1379–1389.
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[16] NASA Space Apps Challenge 2026 Bangladesh Participants Guide — user-provided PDF; especially pp. 9–17 for videos, submission, judging and awards.

ASSUMPTIONS TO RECHECK BEFORE RELEASE

Dataset versions, variable names, parameter availability, temporal coverage, units, spatial support, quality flags, access permissions and download routes can change. Confirm them against current NASA documentation during implementation. Proposed case questions are not scientific findings. This document is a product and build strategy, not a completed analysis.

Prepared as a working master plan for the team. The team should update the implementation status, actual dataset versions, user-test results, public links and final video schedule as the prototype develops.